“Injury Risk Prediction System for Athletes Using Performance Data”

CO2 ASSESSMENT TOOL 1

Submitted in the partial fulfilment for the Course of

CSA1016 - Software Engineering

to the award of the degree of

BACHELOR OF
TECHNOLOGY IN
Information Technology

Submitted by

192521491- DILLI BABU N

Under the
Supervision of Dr. K
Anita Davamani

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical
Sciences Chennai-602105

August 2025

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences Chennai-602105

DECLARATION

I, **Dilli Babu N 192521491,** of the **Department of Information Technology**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the project work entitled **“Injury Risk Prediction System for Athletes Using Performance Data”** is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

Date:

Signature of the Student

Table of Contents

1. Introduction

The Injury Risk Prediction System for Athletes Using Performance Data is a cloud-based, web- and mobile-accessible software system designed to help coaches, sports scientists, and medical staff proactively identify athletes at elevated risk of injury. The system continuously collects objective training-load data (from wearables and manual entry), subjective wellness data, and historical injury records, and applies workload analytics and a machine-learning model to generate a daily injury-risk score for every athlete.

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements, actors, use cases, data requirements, external interfaces, system features, assumptions, and constraints of the proposed platform, following a scenario-based requirement analysis approach.

2. Problem Description

Coaching and medical staff working with athletes and sports teams frequently struggle to identify injury risk before it results in an actual injury. Existing practices (spreadsheets, isolated wearable-vendor apps, and subjective judgment) create serious challenges:

- Training-load data, wellness responses, and injury history are scattered across separate tools and spreadsheets, making it hard to build a complete risk picture for each athlete.
- There is no reliable, automated way to detect dangerous spikes in training load (e.g., acute:chronic workload ratio) before an injury occurs.
- Injury risk assessment currently relies on subjective coach/physio judgment rather than a consistent, data-driven score.
- Athletes' daily wellness (sleep, soreness, fatigue, stress) is rarely captured systematically or correlated with training load.
- When injuries do occur, records of type, severity, and rehabilitation progress are fragmented, making it difficult to learn from past outcomes.
- Coaches and medical staff have no real-time, at-a-glance view of which athletes in a squad currently need attention.

The proposed system addresses these issues by providing a single platform that ingests performance and wellness data, calculates workload metrics, runs a predictive risk model, and surfaces clear, actionable, real-time alerts to the people who can act on them.

3. Scope of the System

The system scope covers the complete athlete-monitoring lifecycle: team and athlete profile setup, performance and wellness data ingestion (wearable and manual), workload-metric calculation (ACWR), machine-learning-based injury-risk scoring, real-time risk dashboards and alerts, injury logging and return-to-play tracking, reporting/export, and role-based access control for every stakeholder.

The system is a web-based application with a companion mobile-friendly interface for athlete wellness entry. It does not cover match tactics/analysis, film/video analysis, contract or financial management, or general electronic health-record (EHR) functionality unrelated to injury-risk monitoring, which lie outside the boundary of this platform.

4. Stakeholders, Business Domain and System Objectives

4.1 Business Domain

The system operates in the sports science and athlete health-management domain, supporting individual athletes, coaching staff, sports medicine teams, and club/organization management across professional, semi-professional, and collegiate sports settings.

4.2 Stakeholders

- Athletes — provide performance and wellness data; primary beneficiaries of reduced injury risk.
- Coaches / Strength & Conditioning Coaches — adjust training load and programs based on risk information.
- Sports Physiotherapists / Team Doctors — review risk alerts, log injuries, manage rehabilitation and return-to-play.
- Sports Scientists / Performance Analysts — monitor workload data and validate incoming data quality.
- Team Managers / Club Management — view read-only summary reports for squad planning (Guest role).
- System Administrators — manage accounts, permissions, and system health.
- Product Owner/Organization — owns the platform and its business goals.

4.3 System Objectives

- Provide a single source of truth for athlete performance, wellness, and injury data.
- Automatically detect dangerous spikes in training load using workload metrics such as ACWR.
- Generate a consistent, explainable, data-driven injury-risk score for every athlete, every day.
- Deliver real-time dashboards and alerts so staff can intervene before an injury occurs.
- Maintain a complete, structured injury history and return-to-play tracking system.
- Ensure secure, role-based access to sensitive athlete health data.

5. Functional Requirements

ID	Functional Requirement	Description
FR-01	User Registration & Login	The system shall allow users to register, log in, and manage their profile using secure authentication (including optional multi-factor authentication).
FR-02	Athlete Profile Creation	The system shall allow a Coach/Admin to create an athlete profile capturing baseline biometrics (age, height, weight, position) and injury-history data.
FR-03	Wearable Data Import	The system shall allow performance data (GPS distance, heart rate, sprint counts) to be imported automatically from connected wearable devices after each session.
FR-04	Manual Session Logging	The system shall allow a Coach to manually log session RPE (Rate of Perceived Exertion) and session details when device data is unavailable.
FR-05	Daily Wellness Questionnaire	The system shall allow an Athlete to submit a daily wellness questionnaire (sleep, soreness, fatigue, stress).
FR-06	Data Validation & Anomaly Flagging	The system shall automatically validate incoming data and flag incomplete or anomalous entries for review before they are used in analysis.
FR-07	Workload Calculation (ACWR)	The system shall automatically calculate each athlete's Acute:Chronic Workload Ratio and flag athletes who exceed a configured threshold.
FR-08	Injury Risk Score Generation	The system shall run a trained machine-learning model on combined performance, wellness, and injury-history data to generate a daily injury-risk score (Low/Moderate/High) per athlete.
FR-09	Risk Factor Explanation	The system shall display the key factors contributing to an athlete's current risk score.
FR-10	Risk Dashboard	The system shall display a color-coded dashboard showing the current risk level of every athlete in a squad.
FR-11	High-Risk Alerts	The system shall automatically notify relevant staff when an athlete's risk score crosses a high-risk threshold.
FR-12	Injury Logging	The system shall allow a Physiotherapist/Team Doctor to log details of an actual injury (type, date, severity, body part) when it occurs.
FR-13	Return-to-Play Tracking	The system shall allow authorized medical staff to record and update an athlete's return-to-play status and rehabilitation milestones.

ID	Functional Requirement	Description
FR-14	Injury History Timeline	The system shall display an athlete's complete injury history in chronological order.
FR-15	Reporting & Export	The system shall allow authorized users to export team risk/workload summaries and raw data as PDF and CSV.
FR-16	Role-Based Access Control	The system shall support roles (Admin, Coach, S&C Coach, Physiotherapist/Doctor, Sports Scientist, Athlete, Guest) with distinct permissions for viewing and editing data.
FR-17	Administration Console	The system shall allow Administrators to manage user accounts, monitor usage, and configure system-wide settings.

6. Non-Functional Requirements

Category	Requirement
Performance	A newly calculated injury-risk score and any resulting high-risk alert shall be delivered to relevant staff within 5 seconds of the daily prediction job completing.
Scalability	The system shall support at least 5,000 concurrently monitored athletes across multiple teams/organizations, with horizontal scaling of application and data-ingestion servers during peak load.
Security	All data in transit shall be encrypted using TLS 1.2+ and data at rest shall be encrypted using AES-256; access to health data shall be governed by role-based permissions.
Reliability	The system shall guarantee 99.9% uptime and shall not lose committed performance, wellness, or injury data even in the event of a server failure (durable storage/backups).
Usability	The dashboard and data-entry interfaces shall be usable by non-technical coaching and medical staff with minimal training, and the athlete wellness form shall be completable on a mobile device in under one minute.
Maintainability	The system shall follow a modular, service-oriented architecture so the data-ingestion, workload-calculation, and ML-prediction services can be updated or retrained independently.
Availability	Core dashboard and alerting features shall remain available 24/7, with scheduled maintenance windows communicated to users in advance.
Portability	The web client shall function consistently across major modern browsers (Chrome, Firefox, Edge, Safari) and responsive mobile layouts.
Auditability	Every risk-score generation, injury record change, and access-control modification shall be logged with user ID, timestamp, and action performed.

7. Actors and Use Cases

7.1 Primary and Secondary Actors

Actor	Type	Description
Athlete	Primary	Submits daily wellness data; views own risk score and injury history.
Coach / S&C Coach	Primary	Manages athlete profiles and training data; views the risk dashboard; adjusts training load.
Physiotherapist / Team Doctor	Primary	Reviews risk alerts and factors, logs injuries, manages return-to-play status.
Sports Scientist / Performance Analyst	Primary	Monitors workload metrics (ACWR), validates data quality, exports reports.
System Administrator	Primary	Manages user accounts, access permissions, and monitors system health.
Team Manager	Secondary	Views shared risk/workload reports in read-only mode; cannot edit (Guest).
Notification Service	Secondary (System Actor)	Automated component that dispatches in-app and email/SMS alerts.

7.2 Use Case Diagram

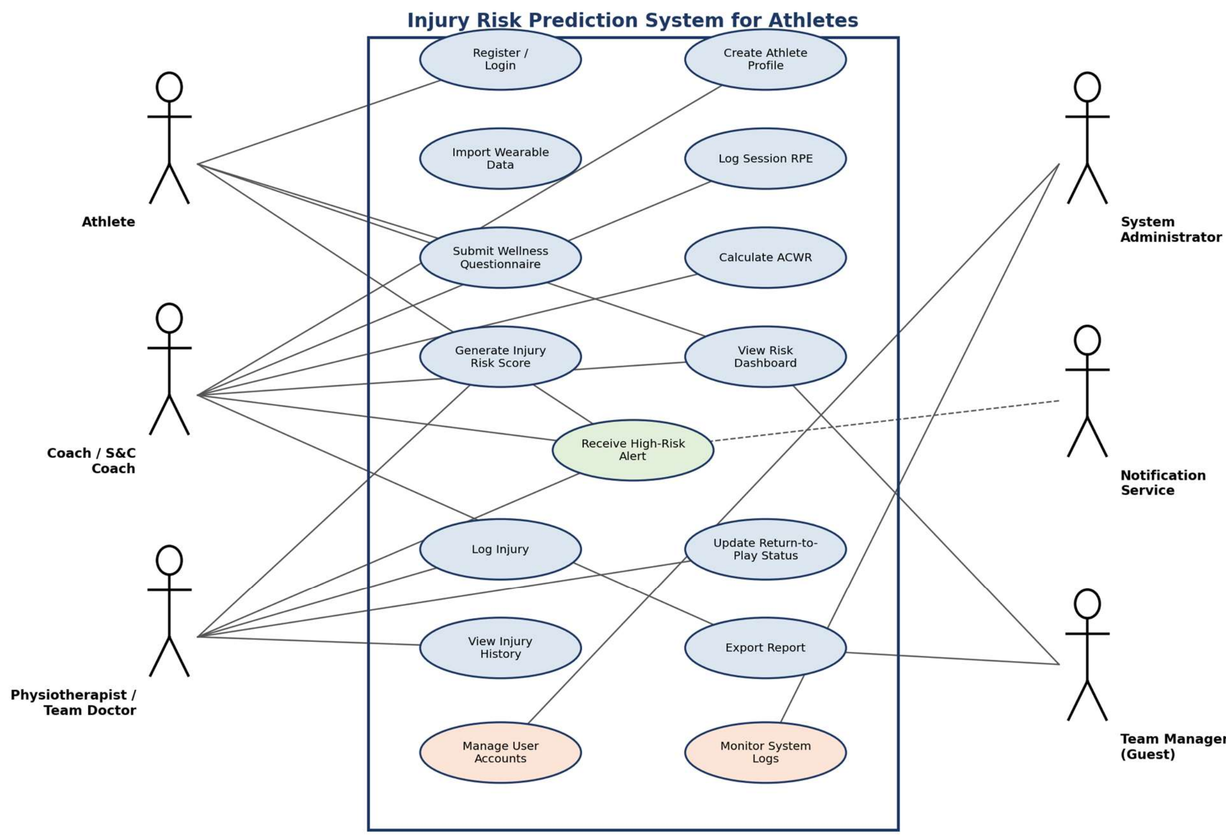

Fig. 1 — Use Case Diagram: Injury Risk Prediction System for Athletes Using Performance Data

7.3 Use Case Descriptions

Use Case	Primary Actor	Description
Register / Login	All Users	Actor creates an account or logs in to access the platform.
Create Athlete Profile	Coach	Actor adds a new athlete with baseline biometric and position details.
Import Wearable Data	System / Sports Scientist	System imports GPS, heart-rate, and sprint data from connected wearables after each session.
Log Session RPE	Coach	Actor manually records session RPE and duration when device data is unavailable.
Submit Wellness Questionnaire	Athlete	Actor submits daily sleep, soreness, fatigue, and stress responses.
Calculate ACWR	System / Sports Scientist	System computes the acute:chronic workload ratio and flags elevated-workload athletes.
Generate Injury Risk Score	System / Data Scientist	System runs the ML model on combined data to produce a daily risk score per athlete.
View Risk Dashboard	Coach / Physiotherapist / Team Manager	Actor views a color-coded overview of all athletes' current risk levels.
Receive High-Risk Alert	Coach / Physiotherapist / Notification Service	Actor is notified automatically when an athlete's risk score crosses a high-risk threshold.
Log Injury	Physiotherapist / Team Doctor	Actor records details of an actual injury against an athlete's profile.
Update Return-to-Play Status	Physiotherapist / Team Doctor	Actor records and updates rehabilitation milestones and availability.
View Injury History	Physiotherapist / Coach	Actor views an athlete's complete chronological injury history.
Export Report	Sports Scientist / Team Manager	Actor exports team risk/workload summaries or raw data as PDF/CSV.
Manage User Accounts	Administrator	Actor creates, disables, or configures roles for user accounts.
Monitor System Logs	Administrator	Actor reviews audit logs and system usage/performance metrics.

8. Data Requirements

- User Data: user ID, name, email, hashed password/auth token, role, profile details.
- Athlete Profile Data: athlete ID, name, date of birth, height, weight, position, team ID, baseline injury history.
- Performance/Training Data: session ID, athlete ID, date, GPS distance, heart-rate metrics, sprint count, session RPE, data source (device/manual).
- Wellness Data: response ID, athlete ID, date, sleep quality, soreness, fatigue, stress score.
- Workload & Risk Prediction Data: ACWR value, risk score, risk level, contributing-factor breakdown, model version, timestamp.
- Injury Records: injury ID, athlete ID, type, date, severity, body part, status, rehabilitation milestones, return-to-play date.
- Access Control Data: role assignments, permission scopes, invitation status per team.
- Notification Data: event type, recipient, delivery channel, read/unread status.
- Audit Log Data: action type, actor, target object, timestamp, IP/session metadata.

9. External Interface Requirements

9.1 User Interfaces

- Responsive web application with a team risk dashboard, athlete profile view, workload/ACWR charts, and injury/return-to-play tracker.
- Mobile-friendly interface for athletes to submit the daily wellness questionnaire.

9.2 Hardware Interfaces

- Wearable training devices (GPS trackers, heart-rate monitors) used during training sessions.
- Standard client devices (desktop, laptop, tablet, smartphone) with a modern web browser and internet connectivity.

9.3 Software Interfaces

- Authentication provider (OAuth 2.0 / SSO integration for institutional or organizational login).
- Wearable-vendor APIs/SDKs for training-load and biometric data import.
- Email/SMS gateway for alert and notification delivery.
- Cloud object storage service for exported reports and attachments.
- ML model-serving service for injury-risk prediction.

9.4 Communication Interfaces

- HTTPS/REST APIs for standard client-server operations.
- WebSocket-based protocol for real-time dashboard updates and alert delivery.

10. System Features

Feature	Priority	Summary
Injury Risk Dashboard	High	Real-time, color-coded view of every athlete's current risk level.
ML-Based Risk Prediction Engine	High	Daily injury-risk score generated from performance, wellness, and injury data.
Wearable & Manual Data Ingestion	High	Automated wearable sync plus manual RPE and wellness entry, with validation.
Workload Analytics (ACWR)	High	Automatic detection of dangerous training-load spikes.
Alerts & Notifications	High	Automatic in-app/email alerts when risk crosses a high-risk threshold.
Injury Logging & Return-to-Play Tracking	High	Structured record of injuries and rehabilitation milestones.
Reporting & Export	Medium	One-click export of risk/workload summaries to PDF and raw data to CSV.
Admin Console	Low	Account management, usage monitoring, and system configuration for administrators.

11. System Architecture (Supporting Diagram)

The following layered architecture diagram illustrates how the client, application, data, and infrastructure layers interact to deliver continuous data ingestion, workload analytics, and ML-based injury-risk prediction.

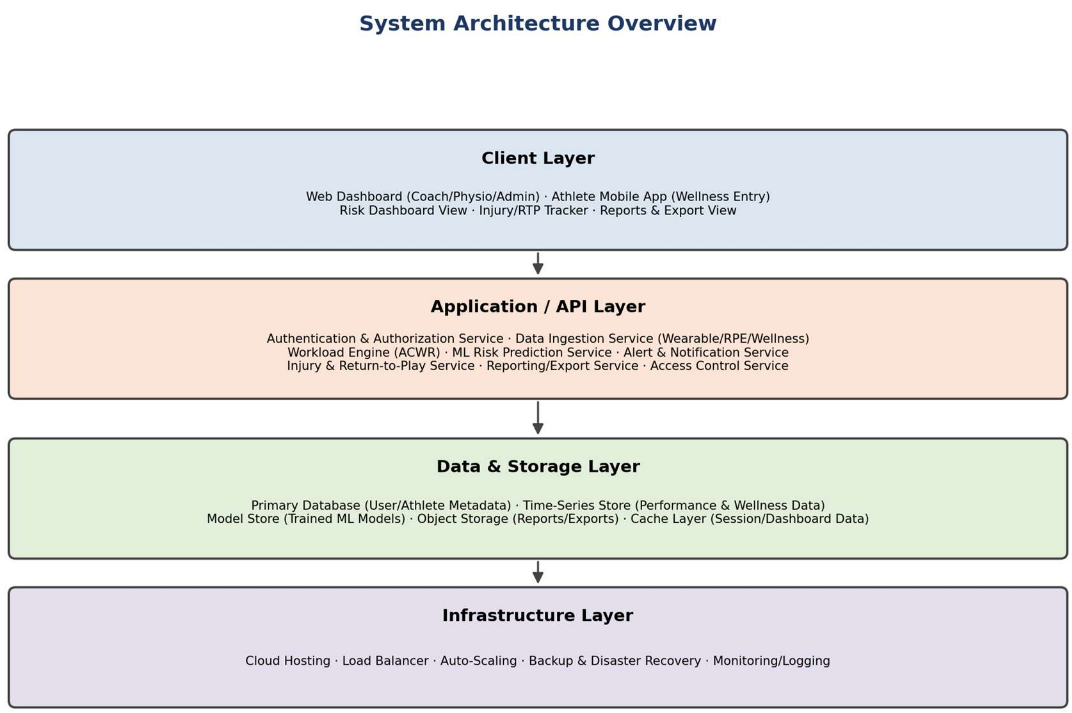

Fig. 2 — High-Level Layered System Architecture

12. Assumptions

- Athletes have access to a wearable device or are willing to have training sessions manually logged by coaching staff.
- Each team has at least one designated Coach or Administrator responsible for managing athlete profiles and staff access.
- Users have consistent access to a stable internet connection while using real-time dashboard and alerting features.
- Athletes and staff are responsible for the accuracy of manually entered data (RPE, wellness responses); the system validates format and range but not truthfulness.
- Third-party wearable-vendor APIs and authentication providers maintain their own agreed service-level agreements (SLAs).

13. Constraints

Type	Constraint
Technical	The ML risk-prediction model requires a sufficient volume of historical performance and injury outcome data before it can produce reliable predictions.
Operational	The system must remain available during training sessions and match days across multiple time zones for touring/travelling teams.
Economic	Cloud storage and compute costs scale with continuous data ingestion and periodic model retraining, requiring a data-retention and retraining-frequency policy.
Legal	The system must comply with health-data protection regulations (e.g., GDPR/HIPAA-equivalent) governing athlete biometric and medical data.
Time-Related	The initial release must prioritize the core data pipeline and workload metrics (ACWR) before the full ML risk-prediction engine, within the project timeline.

14. Requirement Validation — Completeness and Consistency

14.1 Stakeholder Coverage Check

Each identified stakeholder (Athlete, Coach/S&C Coach, Physiotherapist/Team Doctor, Sports Scientist, Team Manager, Administrator) is mapped to at least one functional requirement and one use case, confirming that no stakeholder's core need has been left unaddressed.

14.2 Ambiguity, Redundancy and Inconsistency Review

- Ambiguity: Terms such as 'high-risk threshold' (FR-11) and 'peak load' (Scalability NFR) are flagged as needing quantified values in the next design iteration (e.g., defining the high-risk threshold as a risk score above a specific numeric cut-off).
- Redundancy: FR-07 (Workload Calculation/ACWR) and FR-08 (Injury Risk Score Generation) were reviewed to confirm they serve distinct purposes — ACWR is one input signal, while the ML risk score is the combined prediction — rather than duplicating functionality.
- Inconsistency: Risk-level terminology was standardized across FR-08, FR-10, FR-11, and the Use Case Descriptions to consistently use the levels Low, Moderate, and High.
- Missing Requirements Identified During Review: an explicit data-validation/anomaly-flagging requirement (added under FR-06) and an audit-logging requirement (added under Data Requirements and the Auditability NFR) were included after the initial pass.

14.3 Suggested Improvements

- Quantify vague terms (e.g., define the exact ACWR threshold and the numeric risk-score cut-off for 'High' risk) in the detailed design phase.
- Add explicit acceptance criteria for each functional requirement to support test-case generation.
- Introduce a formal requirement traceability matrix linking each FR/NFR to its corresponding use case and test case.
- Conduct a stakeholder walkthrough/review session with coaching and medical staff before moving to the design phase, to validate assumptions about data-entry workflows.

15. Conclusion

This Software Requirement Specification has analyzed the problem of reactive, fragmented injury management in sports teams and defined a scenario-based set of requirements for an Injury Risk Prediction System. By combining automated performance and wellness data ingestion, workload analytics, and a machine-learning risk-prediction engine with real-time dashboards and alerts, the proposed system gives coaches and medical staff the early warning they need to intervene before an injury occurs. The functional and non-functional requirements, actor and use-case models, data and interface requirements, and the validation review in this document together provide a complete and consistent foundation for the subsequent design and Agile sprint-planning phases of the project.